

Chemistry, 2018 EC, entrance exam, version 4

Grade 12 · Chemistry · 2018 E.C.

Time allowed: 120 minutes**Questions:** 75**Total marks:** 75

Name _____ Grade _____ ID No. _____ Date _____

1. Modern atomic theory contradicts Dalton's atomic theory in stating that: (1)
- A. When elements combine, their atoms unite in small whole number ratio to form compound.
 - B. All substances are made up of very small particles called atoms.
 - C. Atoms of different elements have different properties.
 - D. Atom can split into smaller particles.
-
2. The atomic nucleus was discovered with the help of an experiment called:_____. (1)
- A. The deflection of cathode ray experiment.
 - B. The Alpha particles scattering experiment.
 - C. The discharge tube experiment.
 - D. The oil drop experiment.
-
3. Which experiment provide evidence that atom is mostly empty space and led to the discovery of the atomic nucleus? (1)
- A. Millikan's oil drop experiment
 - B. Bohr's hydrogen atom experiment
 - C. Rutherford's gold foil experiment
 - D. Thomson's cathode ray tube experiment
-
4. An element has electron configuration of $[\text{Kr}]5s^24d^4$. To which group and period respectively does the element belong in the Modern periodic table? (1)
- A. Group IVB and period 4
 - B. Group IIB and period 5
 - C. Group VB and period 6
 - D. Group VIB and period 5
-
5. The characteristic similarities of elements in the same period is: (1)
- A. They all have the same electron configuration
 - B. They all have the same valence electrons
 - C. They all have the same number of energy level
 - D. They all have similar chemical properties
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6. Which One of the following statement is not true about Mendeleev's periodic table? (1)
- A. Arranged elements in order of increasing atomic masses
 - B. It was proposed in 1869 and predicted many new elements
 - C. Wrong order of atomic masses would be observed
 - D. The position of isotopes and noble gases are available
-
7. According to modern periodic law, the periodic properties of elements are periodic functions of their_____. (1)
- A. Atomic sizes
 - B. Atomic numbers
 - C. Atomic masses
 - D. Mass numbers

8. An element 'A' is found in group IIIA and an element 'B' is found in group VIA. The compound formed with these two elements will have the formula:

A. A_2B_6 B. A_6B_2
C. A_2B_3 D. A_3B_2

9. The atomic number of elements Na, Cl, Al and K are 11, 17, 13 and 19 respectively. Which of the following statement is not true?

A. The ionization potential of K is smaller than that of Na
B. The electron affinity of Cl is smaller than that of Al
C. Chlorine has the highest effective nuclear charge than that of K
D. The metallic character of K is greater than that of Na

10. Out of the elements ${}_7\text{N}$, ${}_{11}\text{Na}$, ${}_{17}\text{Cl}$, and ${}_{15}\text{P}$:

A. N, Na and P are found in the same period
B. Cl and N have the same number of energy level
C. N and P are found in the same group
D. Na and Cl are found in the same group

11. Which of the following is not the advantages of periodic classification of elements in studying chemistry?

A. It helps to predict physical and chemical properties, reactivities, and types of compound form.
B. It clearly show trends in properties of elements across a periods and down the groups.
C. It simplifies the study of elements by grouping those with similar chemical behavior.
D. It arranges isotopes of an element in different location.

12. What are the real cause of chemical bonding?

A. Increasing the potential energy of unstable atoms
B. Complete the transfer of electrons from one atom to another.
C. Attending stable electron configurations.
D. Sharing electrons between atom of different elements.

13. Which one of the following is the characteristics of most ionic compounds?

A. They are generally liquid or gases at ordinary temperature.
B. They conduct electric current when in molten or aqueous solution.
C. They have relatively low melting and boiling points.
D. They are soluble in non polar solvents

14. Covalent bonds are typically formed between atoms with:

A. High ionization energy and high electron negativity
B. Low ionization energy and high electron negativity
C. High ionization energy and low electron negativity
D. Low ionization energy and low electron negativity

15. Which one of the following compounds represent atoms bonded covalently? 1. N_2O_2 2. NH_3 3. MgBr_2 4. BaO 5. SO_3

A. 1, 3 and 5 B. 1, 2 and 5
C. 3 and 4 D. 1, 2 and 3

16. A type of chemical bonding that results from the mutual attraction between positively charged metal nuclei and the surrounding freely moving sea of electrons is named as: (1)
- A. Covalent
B. Dative
C. Ionic
D. Metallic
17. What type of intermolecular force is responsible for high boiling of water? (1)
- A. Dipole-dipole interaction
B. Hydrogen bonding
C. London or dispersion forces
D. Covalent bonding
18. The correct balanced chemical equation for the reaction: Phosphorus + oxygen $\rightarrow$ phosphorus(V) oxide is: (1)
- A. $P_4 + 5O_2 \rightarrow P_4O_{10}$
B. $P_5 + O_2 \rightarrow P_2O_5$
C. $5P + O_2 \rightarrow P_5O_2$
D. $2P + O_2 \rightarrow 2PO$
19. Chemical equation must be balanced to satisfy which one of the following law? (1)
- A. Conservation of energy
B. Definite proportion
C. Conservation of mass
D. Multiple proportion
20. Which one of the following is an example of a single displacement reaction? (1)
- A. $CaO(s) + SiO_2 \rightarrow CaSiO_3(s)$
B. $2NaHCO_3(aq) \rightarrow Na_2CO_3(s) + CO_2(g) + 2H_2O(l)$
C. $2KI(aq) + F_2(g) \rightarrow 2KF(aq) + I_2(g)$
D. $H_2SO_4(aq) + 2KOH(aq) \rightarrow K_2SO_4(aq) + 2H_2O(l)$
21. Which of the following statements is correct? (1)
- A. The oxidizing agent is always oxidized in oxidation and reduction reaction
B. The reducing agent is always reduced in oxidation and reduction reaction
C. The oxidizing agent in oxidation and reduction reaction always loses electrons
D. A reducing agent in oxidation and reduction reaction always loses electrons
22. Calculate the formula mass of hydrated sodium carbonate ($Na_2CO_3 \cdot 10H_2O$) (atomic mass of Na=23, C=12, O=16, H=1) (1)
- A. 326
B. 286
C. 274
D. 142
23. Which solute particles dispersed in water are increasing in size in a mixture? (1)
- A. Suspension, Colloid, Solution
B. Solution, Suspension, Colloid
C. Colloid, Solution, Suspension
D. Solution, Colloid, Suspension
24. How can you best increase the solubility of a gas in liquid? (1)
- A. Increasing both the temperature and the pressure
B. Decreasing both the temperature and the pressure
C. Increasing the temperature and decreasing the pressure
D. Decreasing the temperature and increasing the pressure
25. In which of the following case will the dissolution of sugar be the most rapid? (1)
- A. Powdered sugar in cold water
B. Sugar crystal in hot water
C. Powdered sugar in hot water
D. Sugar crystal in cold water

26. The solubility of NaCl in water at 20°C is 40g in 100g of water. A solution that contains 20g of NaCl in 50g of water at 20°C is _____. (1)
- A. Unsaturated
B. Saturated
C. Super saturated
D. Diluted
27. How many grams of glucose (C₆H₁₂O₆) must be dissolved in 660g of ethanol (C₂H₅OH) to prepare a 4.40 × 10⁻⁴ molality of solution? (1)
- A. 0.560g
B. 0.0522g
C. 2.430g
D. 3.661g
28. A solution is said to be in equilibrium if and only if: (1)
- A. It forms suspension
B. It dissolves more solutes
C. Its rate of dissolution is equal to the rate of crystallization
D. Solute-solvent interaction is less than that of solute-solute interaction
29. What is the concentration of sodium hydroxide required to react completely with equal volume of 0.5M of phosphoric acid solution? If 0.25L of H₃PO₄ is used? $\text{H}_3\text{PO}_4(\text{aq}) + 3\text{KOH}(\text{aq}) \rightarrow \text{K}_3\text{PO}_4 + 3\text{H}_2\text{O}$ (1)
- A. 0.375M
B. 0.125M
C. 1.5M
D. 2.5M
30. What is the mass percentage of methanol in a solution prepared by adding 32g of methanol to 18g of water? (1)
- A. 16%
B. 64%
C. 50%
D. 32%
31. A compound formed when an acidic oxide react with a basic oxide is: (1)
- A. Acid
B. Salt
C. Base
D. Oxide
32. Which of the following statement is true about oxide? (1)
- A. All basic oxides are metallic oxides
B. Amphoteric oxides are basic oxides
C. All non-metallic oxides are acidic oxides
D. Oxides are ternary compounds
33. What is the concentration of hydrogen ions in 0.0001 M NaOH aqueous solution? (1)
- A. 2.0×10^{-4} M
B. 1.0×10^{-10} M
C. 1.0×10^{-8} M
D. 1.0×10^{-14} M
34. Which one of the following acid is used as electrolyte in a car battery? (1)
- A. Hydrochloric acid
B. Nitric acid
C. Sulphuric acid
D. Phosphoric acid
35. The idea of atoms as the smallest particles of substances was proposed by: (1)
- A. James Chadwick
B. John Dalton
C. Ernest Rutherford
D. J.J Thomson

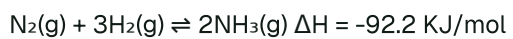
36. A sample of water taken from ice, rain or ocean has the same ratio by mass of hydrogen and oxygen. This chemical fact explain the law of: (1)
- A. Mass action
B. Conservation of mass
C. Definite proportion
D. Multiple proportion
37. All cathode rays are the same regardless of the nature of material used as electrode within the discharge tube. This observation indicates that: (1)
- A. Cathode rays are energetic
B. Cathode rays travel in a straight line
C. Cathode rays are electrically charged
D. Cathode rays are building block of all matter.
38. Calculate RAM for naturally occurring Mg if the isotope distribution is 78.99% ^{24}Mg , 10.00% ^{25}Mg and 11.01% ^{26}Mg ; accurate masses are 23.99, 24.99 and 25.98 respectively. (1)
- A. 26.51
B. 24.31
C. 22.41
D. 25.35
39. Which of the following electron transitions requires the smallest energy to be absorbed by the hydrogen atom? (1)
- A. from $n = 1$ to $n = 2$
B. from $n = 2$ to $n = 3$
C. from $n = 3$ to $n = 4$
D. from $n = 4$ to $n = 5$
40. The normal human eye responds to visible light of wavelength ranging from about 390nm to 710nm. What is the frequency range of the human eye? (1)
- A. 7.7×10^{14} Hz to 4.2×10^{14} Hz
B. 4.2×10^{14} Hz to 7.7×10^{14} Hz
C. 1.3×10^{-15} Hz to 2.4×10^{-15} Hz
D. 2.4×10^{-15} Hz to 1.3×10^{15} Hz
41. Which of the following statements is TRUE? (1)
- A. Ultraviolet light has longer wavelength than visible light
B. The energy of radiation decreases as the wavelength decreases
C. The frequency of radiation increases as the wavelength decreases
D. Wave number of an electromagnetic radiation increases as wavelength increases
42. Which of the following set of quantum numbers is permissible? (1)
- A. $n=3, l=3, m_l=-2, m_s=-1/2$
B. $n=2, l=0, m_l=-1, m_s=+1/2$
C. $n=2, l=0, m_l=0, m_s=+1/2$
D. $n=0, l=0, m_l=1, m_s=-1/2$
43. An element has electron configuration of $[\text{Kr}]5s^24d^{10}5p^5$. To which group and period respectively does the element belong in the Modern periodic table? (1)
- A. Group IVA and period 4
B. Group VA and period 7
C. Group IIA and period 5
D. Group VIIA and period 5
44. Which one of the following ionic compound has the highest lattice energy? (1)
- A. NaCl
B. NaBr
C. NaF
D. NaI
45. Which of the following compound contain ionic, covalent and coordinate covalent bond all in one? (1)
- A. KCl
B. MgCO_3
C. NH_4Cl
D. HCOOH

46. Which of the following molecule is polar covalent molecule? (1)
- A. CO_2 B. CCl_4
C. PCl_5 D. NH_3
47. The species in the following set do not include which one of the following geometry? SiCl_4 , BrF_4^- , SF_6 , NO_3^- (1)
- A. Octahedral B. Tetrahedral
C. Trigonal pyramidal D. Square planar
48. The type of hybridization of the central atom in ClF_3 and SF_6 are __ and __ respectively? (1)
- A. sp^3d and sp^3 B. sp^3d and sp^3d^2
C. sp^3 and sp^3d D. sp^2 and sp^3d^2
49. Which of the following statement is incorrect? (1)
- A. A molecule may be nonpolar, although all the bonds in it are polar
B. If a molecule BF_3 is nonpolar, the orbitals on B are sp^3 hybridized
C. In a crystal ammonium bromide there are ionic and covalent bond
D. In ammonium ion NH_4^+ , all the bond lengths are identical
50. In which of the following molecules the dispersion or London force is the most constant? (1)
- A. CH_3Cl and CO B. Ar and CCl_4
C. H_2O and NH_3 D. HCl and CHCl_3
51. As atoms bonded to each other, they: (1)
- A. Decrease their potential energy, thus creating less stable arrangement of matter.
B. Decrease their potential energy, thus creating more stable arrangement of matter.
C. Increase their potential energy, thus creating less stable arrangement of matter.
D. Increase their potential energy, thus creating more stable arrangement of matter.
52. Metallic bonding is formed between: (1)
- A. Elements of low electronegativities B. Elements of small atomic sizes
C. Elements of high electronegativities D. Elements of high electron affinities
53. Which of the following molecule or ion is least stable? (1)
- A. CN B. CN^-
C. O_2 D. F_2
54. According to molecular orbital theory which one of the following is paramagnetic? (1)
- A. NO B. F_2
C. NO^+ D. Be_2
55. Identify the wrong statement from the following alternatives: (1)
- A. F_2 molecule is diamagnetic
B. The bond order of N_2 is 3
C. The bond energy of CN^- is less than that of O_2
D. The bond length of F_2 is greater than that of N_2

56. Molecules are held together in a crystal by: (1)
- A. Hydrogen bonds
B. Van der Waal's attraction
C. Electrostatic attraction
D. Dipole-dipole attraction
57. At high temperatures the particles of a gas: (1)
- A. Move slower and hit the walls of their container less often.
B. Move faster and hit the walls of their container more frequently
C. Move faster and hit the walls of their container less frequently
D. Don't move at all and don't hit the walls of their container.
58. According to the kinetic-molecular theory, particles of a gas (1)
- A. Attract each other but they do not collide
B. Repel each other and they collide
C. Neither attract nor repel each other but they collide
D. Neither attract nor repel each other and they do not collide
59. Which one of the following statements is NOT true about acids? (1)
- A. Brønsted-Lowry acid is a proton donor
B. Lewis acid is an electron pair donor
C. Lewis acid is an electron pair acceptor
D. Arrhenius acid furnishes hydronium ion in aqueous solution
60. Which of the following is the conjugate acid of hydrogen phosphate ion, (HPO_4^{2-})? (1)
- A. H_3PO_4
B. PO_4^{3-}
C. H_2PO_4^-
D. HPO_4^{2-}
61. What is the pH of a 5×10^{-6} M solution of $\text{Ba}(\text{OH})_2$? (1)
- A. 9
B. 11
C. 12
D. 10
62. A 1.0M solution of a weak acid is found to dissociate by only 1.4%. Which of the following acid is it likely to be: (1)
- A. HNO_2 , $K_a = 6.91 \times 10^{-4}$
B. HCOOH , $K_a = 1.96 \times 10^{-4}$
C. HF , $K_a = 7.2 \times 10^{-4}$
D. CH_3COOH , $K_a = 1.82 \times 10^{-5}$
63. If KOCCl is added to a solution of HOCl , which of the following statements is true? (1)
- A. The pH of the solution will increase
B. The pH of the solution will decrease
C. The equilibrium will not be affected
D. The pH will remain the same
64. Buffer solution have consistent acidity and alkalinity because: (1)
- A. They have fixed value of pH
B. They have excess of H^+ or OH^- ions
C. The acids and alkalies are shielded from attack by other ions
D. They give unionized acids or bases on the reaction with added acid or base
65. Which of the following salts will yield a basic solution on dissolution in water? (1)
- A. A salt of weak acid and weak base
B. A salt of weak acid and strong base
C. A salt of strong acid and strong base
D. A salt of strong acid and weak base

66. What molarity of 50mL H_2SO_4 is required to neutralize 200mL of 0.5M NaOH? (1)
- A. 1M
B. 4N
C. 2M
D. 0.5M
67. Which of the following titration will have an equivalent point of pH less than 7? (1)
- A. Strong acid with weak base
B. Weak acid with weak base
C. Strong acid with strong base
D. Weak acid with strong base
68. For the reaction: $\text{H}_2\text{S} + \text{HNO}_3 \rightarrow \text{S} + \text{NO} + \text{H}_2\text{O}$, what is the coefficient H_2S and NO respectively after balancing redox reaction by oxidation number method? (1)
- A. 2 and 4
B. 3 and 2
C. 3 and 4
D. 4 and 2
69. Consider the following oxidation-reduction reaction equation: $2\text{H}_2\text{O}(\text{l}) + \text{Al}(\text{s}) + \text{MnO}_4^-(\text{aq}) \rightarrow \text{Al}(\text{OH})_4^-(\text{aq}) + \text{MnO}_2(\text{s})$. What are the reducing and the oxidizing agents in this reaction? (1)
- A. Al(s) is the reducing agent and H_2O is the oxidizing agent
B. Al(s) is the oxidizing agent and H_2O is the reducing agent
C. Al(s) is the oxidizing agent and $\text{MnO}_4^-(\text{aq})$ is the reducing agent
D. Al(s) is the reducing agent and $\text{MnO}_4^-(\text{aq})$ is the oxidizing agent
70. The anode reaction is the same in all of the following electrolysis process except: (1)
- A. Electrolysis of dilute NaCl solution
B. Electrolysis of conc. HCl solution
C. Electrolysis of brine solution
D. Electrolysis of molten MgCl_2
71. Which of the following conduct electricity without undergoing any chemical change? (1)
- A. Potassium chloride solution
B. Hydrochloric acid solution
C. Molten magnesium chloride
D. Copper metal
72. Two cells are connected in series. One cell contains Ag_2SO_4 and the other cell contains $\text{Al}(\text{NO}_3)_3$. If 12g of Al is deposited on one cell, how many grams of Ag is deposited on the other cell? (Atomic mass of Ag = 108g/mole and Al = 27g/mole) (1)
- A. 144g
B. 24g
C. 12g
D. 36g
73. The two standard electrode potentials involved in the nickel-cadmium rechargeable cell are given below. Calculate the ΔG° of the cell. ($F = 96500 \text{ C}$) $\text{NiO}_2(\text{s}) + 2\text{H}_2\text{O}(\text{l}) + 2\text{e}^- \rightarrow \text{Ni}(\text{OH})_2(\text{s}) + 2\text{OH}^-$ $E^\circ = +0.49\text{V}$; $\text{Cd}(\text{OH})_2(\text{s}) + 2\text{e}^- \rightarrow \text{Cd}(\text{s}) + 2\text{OH}^-(\text{aq})$ $E^\circ = -0.76\text{V}$ (1)
- A. -184.6 KJ
B. -153.0 KJ
C. -241.3 KJ
D. -206.8 KJ
74. Which of the following is not the characteristic of Chemical industries? (1)
- A. Consume large quantities of energy during manufacturing process.
B. Use safe operation method during manufacturing process
C. Test their product during and after manufacturing
D. Providing transport services to customers

75. In the production of ammonia, which condition would increase the yield of ammonia according to this reaction? (1)



A. High temperature and low pressure

B. High temperature and high pressure

C. Low temperature and high pressure

D. Low temperature and low pressure

— END OF EXAMINATION —

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